

Mathematical Viability of Coarse-Graining Invariant Set Theory to Bohmian Hydrodynamics: A Rigorous Assessment

Ontological Architecture of Invariant Set Theory and the Coarse-Graining Formulation

Invariant Set Theory (IST), formulated by Tim Palmer, provides a locally causal, deterministic, ψ -epistemic foundation for quantum phenomena grounded in non-linear dynamical systems theory and number theory¹. Rather than defining physical evolution via differential equations on a smooth space-time manifold, IST posits that the universe U is an isolated dynamical system whose trajectories evolve on a measure-zero, fractal invariant subset I_U embedded within a high-dimensional cosmological state space X_U ¹. The geometric structure of I_U is constructed from Cantor sets of trajectories homeomorphic to the algebraic ring of p -adic integers $\mathbb{Z}_p$, where p is a large prime number or large integer parameter ($p \gg 0$)¹. Standard quantum mechanics, formalized over continuous complex Hilbert spaces $\mathcal{H}$, emerges strictly as a singular limit of IST as $p \rightarrow \infty$ or $N \rightarrow \infty$ ¹.

Bohmian mechanics offers an alternative ontology wherein non-relativistic quantum systems consist of point particles advected along deterministic trajectories in configuration space $\mathbb{R}^3$ ⁵.

The dynamics are dictated by a single-valued, continuous velocity field $v_B(x, t)$:

$$v_B(x, t) = \frac{J(x, t)}{\rho(x, t)} = \frac{\hbar}{m} \text{Im} \left(\frac{\nabla \psi(x, t)}{\psi(x, t)} \right) = \frac{\nabla S(x, t)}{m}$$

where $J(x, t)$ is the quantum probability current density, $\rho(x, t) = |\psi(x, t)|^2$ is the Born-rule probability density, and $S(x, t)$ represents the scalar phase of the complex wave function $\psi(x, t) = \sqrt{\rho(x, t)} e^{iS(x, t)/\hbar}$ ⁶.

The primary mathematical objective is to investigate whether a measure-preserving coarse-graining pushforward map $\pi_* : (I_U, T_t, \mu_I) \rightarrow (M, \Phi_t, \rho_B dx)$ can rigorously map the microscopic, discrete trajectory dynamics on the fractal invariant set I_U to the continuous Bohmian velocity field $v_B(x, t)$ and probability continuity equations on a spatial manifold M ¹. Evaluating this micro-to-macro transition requires addressing five fundamental mathematical obstructions: fractal non-rectifiability, kinetic velocity dispersion, Wallstrom circulation quantization, Sinai-Ruelle-Bowen (SRB) measure dissipation, and the kinematic emergence of the quantum potential Q .

Fractal Non-Rectifiability and Differential Calculus Obstructions

Metric Structure of I_U and Derivation Degeneracy

Constructing a continuous velocity field $v = \frac{dx}{dt}$ directly on I_U requires establishing a differential structure on the underlying metric measure space (I_U, d_p, μ_I) , where d_p denotes the p -adic metric and μ_I is the invariant natural measure¹. In non-smooth mathematical analysis—developed by Cheeger, Weaver, Ambrosio, Kirchheim, and Gigli—vector fields on general metric measure spaces (X, d, m) are formalized via Weaver metric derivations⁹.

A Weaver derivation b on a metric measure space (X, d, m) is a linear operator $b : \text{Lip}_b(X, d) \rightarrow L^0(X, m)$ satisfying the Leibniz rule $b(fg) = fb(g) + gb(f)$ and a locality condition $|b(f)|(x) \leq g(x) \cdot \text{lip}_a(f)(x)$ m -almost everywhere, where $\text{lip}_a(f)(x)$ represents the asymptotic Lipschitz constant⁹. The space of p -integrable Lipschitz derivations $\text{Der}^p(X, d, m)$ serves as the functional analytic analogue to the space of vector fields on smooth manifolds⁹.

The topological structure of I_U at finite p is homeomorphic to the p -adic integers $\mathbb{Z}_p$, which possesses a totally disconnected, ultrametric topology¹. In a totally disconnected metric space, every connected component is a single point, implying that the only continuous curves $\gamma : [0, 1] \rightarrow I_U$ are constant maps¹. According to Cheeger's differentiability theorem and the Ambrosio-Kirchheim theory of metric currents, the existence of non-trivial Lipschitz derivations or rectifiable 1-currents requires the metric measure space to be countably

1-rectifiable¹². A metric space is 1-rectifiable if it can be covered m -a.e. by a countable union of Lipschitz images of subsets of $\mathbb{R}^{12}$.

Because the p -adic metric space (I_U, d_p) contains zero non-trivial rectifiable curves, any metric current $T \in \mathcal{M}_1(I_U)$ or derivation $b \in \text{Der}(I_U, d_p, \mu_I)$ computing directional spatial derivatives vanishes identically:

$$\text{Der}_{\text{spatial}}(I_U, d_p, \mu_I) \equiv \{0\}$$

Implications for Microscopic Velocity Fields

The mathematical degeneracy of spatial derivations at finite p proves that a continuous Bohmian velocity field $v_B(x, t) = \frac{\nabla S}{m}$ cannot be extracted via a direct spatial smoothing or local Eulerian coarse-graining of I_U ¹. The differential structure necessary to define velocity vector fields does not exist at the microscopic level $p < \infty$ ¹. Consequently, the macroscopic Bohmian velocity field cannot represent a spatial averaging of microscopic tangent vectors on I_U . Instead, the spatial tangent module $L^2(TM)$ and smooth vector fields emerge extrinsically in the singular limit $p \rightarrow \infty$, wherein the zero-dimensional p -adic topology collapses and the dense projection onto $\mathbb{R}^n$ restores continuous differential calculus¹.

Kinetic Closure, Velocity Dispersion, and Multistreaming Obstructions

Semiclassical Limits and the Hydrodynamic Transition

Transitioning from deterministic microscopic trajectories to macroscopic hydrodynamics requires passing through kinetic transport theory¹⁸. Let $f(x, v, t)$ represent the phase-space density of an ensemble of trajectories on I_U projected onto spatial coordinates $x \in \mathbb{R}^3$ and operational velocities $v \in \mathbb{R}^3$ ¹⁸. The kinetic transport of $f(x, v, t)$ is governed by the Wigner-Vlasov transport equation¹⁸:

$$\partial_t f + v \cdot \nabla_x f + F_{\text{eff}}[f] \cdot \nabla_v f = 0$$

Macroscopic hydrodynamic variables are obtained by taking velocity moments of $f(x, v, t)$ ¹⁸.

$$\rho(x, t) = \int_{\mathbb{R}^3} f(x, v, t) dv$$

$$u(x, t) = \bar{v}(x, t) = \frac{1}{\rho(x, t)} \int_{\mathbb{R}^3} v f(x, v, t) dv$$

Integrating the kinetic transport equation over velocity space yields the spatial continuity equation and the momentum conservation equation⁸:

$$\partial_t \rho + \nabla \cdot (\rho u) = 0$$

$$\partial_t (\rho u_i) + \sum_j \partial_{x_j} (\rho u_i u_j) + \sum_j \partial_{x_j} \mathbb{P}_{ij} = -\rho \partial_{x_i} V$$

where $\mathbb{P}_{ij}(x, t)$ is the kinetic stress tensor (velocity dispersion tensor), defined as¹⁸:

$$\mathbb{P}_{ij}(x, t) = \int_{\mathbb{R}^3} (v_i - u_i)(v_j - u_j) f(x, v, t) dv$$

Monostreaming Failure and Kinetic Closure

The exact Bohmian/Madelung hydrodynamics strictly demands a monostreaming regime (zero local velocity dispersion)⁶:

$$f(x, v, t) = \rho(x, t) \delta^3(v - v_B(x, t)) \implies \mathbb{P}_{ij}(x, t) \equiv 0$$

Under monostreaming, the velocity dispersion vanishes, closing the momentum equation cleanly into the single-velocity quantum Euler equation⁶.

However, the chaotic non-linear dynamics on I_U feature trajectory stretching, folding, and dense fractal clustering¹. Projecting a fractal ensemble of deterministic trajectories from

$I_U \subset X_U$ down to physical spatial coordinates $\mathbb{R}^3$ causes trajectories belonging to different leaves of the Cantor set to cross in configuration space¹. This phenomenon generates multistreaming (caustic formation), wherein multiple distinct trajectory velocities

$v_1, v_2, \dots, v_k$ coexist at a single spatial point x ¹⁸.

As demonstrated by François Golse, Lions, Paul, and Markowich in semiclassical kinetic limits, whenever multistreaming occurs, the velocity dispersion tensor is strictly non-zero ($\mathbb{P}_{ij} \neq 0$)¹⁸. Consequently, the macroscopic momentum equation cannot be expressed solely in terms

of a single-valued field $u(x, t) = v_B(x, t)$, and attempting to force a single-valued velocity field without accounting for $\mathbb{P}_{ij}$ creates an unclosed kinetic hierarchy⁶.

Resolution via Quantum Stress Identification

A mathematically valid bridge from microscopic IST dynamics to Bohmian hydrodynamics exists if and only if the kinetic stress tensor generated by trajectory dispersion on I_U matches the non-local quantum stress tensor $\mathbb{T}_{ij}^{\text{quant}}$ ⁸:

$$\mathbb{P}_{ij}(x, t) = \mathbb{T}_{ij}^{\text{quant}}(x, t) = -\frac{\hbar^2}{4m^2} \rho(x, t) \partial_i \partial_j \ln \rho(x, t)$$

When this identity holds, the divergence of the stress tensor simplifies to the gradient of the quantum potential $Q(x, t)$ ⁷:

$$\frac{1}{\rho} \sum_j \partial_{x_j} \mathbb{T}_{ij}^{\text{quant}} = \partial_{x_i} \left(-\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} \right) = \nabla_i Q$$

Thus, the velocity dispersion obstruction is qualified: coarse-graining IST trajectories does not yield pure mono-kinetic classical fluid flow; rather, the fractal multi-velocity dispersion on I_U must re-emerge macroscopically as the quantum stress tensor driving Q ⁸.

The Wallstrom Circulation Quantization Objection

Wallstrom's Negative Verdict on Hydrodynamic Formulations

In 1989, Timothy Wallstrom identified a critical mathematical flaw in deriving Schrödinger wave mechanics from hydrodynamic/Bohmian equations⁶. The complex Schrödinger equation

implies that the phase $S(x, t)$ is the argument of a single-valued complex scalar field

$\psi(x, t) = \sqrt{\rho(x, t)} e^{iS(x, t)/\hbar}$ ⁶. This topological structure forces the circulation of the velocity field $v = \frac{\nabla S}{m}$ around any closed loop γ in the non-zero density region $\Omega_+ = \{x \in \mathbb{R}^3 : \rho(x, t) > 0\}$ to satisfy the integer quantization condition⁶:

$$\oint_{\gamma} v \cdot dx = \frac{1}{m} \oint_{\gamma} \nabla S \cdot dx = \frac{2\pi\hbar n}{m}, \quad n \in \mathbb{Z}$$

Wallstrom pointed out that the classical Euler-Madelung hydrodynamic equations:

$$\partial_t \rho + \nabla \cdot (\rho v) = 0, \quad m(\partial_t v + (v \cdot \nabla)v) = -\nabla(V + Q)$$

are strictly defined on real-valued field variables (ρ, v) ⁶. On multiply connected domains Ω_+ (created by nodal lines where $\rho = 0$), these hydrodynamic equations admit smooth stationary solutions with arbitrary, non-quantized circulation values $C \in \mathbb{R}$ ⁶. Because the hydrodynamic equations do not intrinsically enforce $C \in \frac{2\pi\hbar}{m}\mathbb{Z}$, Madelung hydrodynamics is mathematically broader than, and not equivalent to, Schrödinger wave mechanics⁶.

Topological Decomposition via Target Regularity and Scalar Descent

Work by Maïke Reddiger and Bill Poirier resolved Wallstrom's objection by proving that the Wallstrom condition decomposes into two independent mathematical requirements⁶:

1. **Target Space Metric Regularity:** The Madelung gradient energy endows the polar value space (r, S) , where $r = \sqrt{\rho}$, with the flat metric $g = dr^2 + \frac{r^2}{\kappa^2} dS^2$, where $\kappa = \sqrt{8m\alpha}$ and α is the Fisher information coefficient⁶. Metric completion at $r = 0$ adds a phase-blind zero-amplitude point. Requiring the completed value space target to be a smooth Riemannian surface uniquely fixes the period of $S/\hbar$ to 2π , establishing $\kappa = \hbar$ and identifying the value space with the complex plane $\mathbb{C}$ ⁶.
2. **Topological Scalar Descent:** Once the target space is identified with $\mathbb{C}$, Madelung field data (ρ, v) yield a globally single-valued complex wave function $\psi = \sqrt{\rho} e^{iS/\hbar}$ if and only if the local phase 1-form $p = dS$ obeys global scalar descent⁶. Under scalar descent, integer circulation $\oint_{\gamma} dS \in 2\pi\hbar\mathbb{Z}$ follows rigorously from topological

degree theory⁶.

Application to Invariant Set Theory

In Invariant Set Theory, state orientations and phase angles ϕ/π are constrained by number-theoretic rules to rational dyadic fractions¹:

$$\frac{\phi}{\pi} \in \mathbb{Q}_2(N) = \left\{ \frac{a}{2^N} : a \in \mathbb{Z} \right\}$$

This discrete phase structure at finite N (or finite p) directly violates continuous single-valuedness¹.

For a coarse-graining bridge to evade Wallstrom's objection, the microscopic trajectory mapping on I_U must not merely yield local velocity fields $v(x, t)$ on Ω_+ ⁶. It must satisfy the global scalar descent condition, ensuring that the macroscopic distribution function pushes forward to a smooth pullback $\psi^* \in C^\infty(M, \mathbb{C})$ ⁶. This descent is mathematically guaranteed in IST only in the singular limit $p \rightarrow \infty$, where dyadic phase angles become dense in $\mathbb{S}^1$, restoring single-valued complex wavefunctions¹.

Reconciling Dissipative Attractors with Conservative Hydrodynamics

Phase-Space Contraction vs. Measure Preservation

In Invariant Set Theory, the cosmological state evolution is modeled as a chaotic, dissipative dynamical system¹. In non-conservative systems, phase-space volume contracts under the flow T_t , characterized by a negative vector field divergence $\nabla_{X_U} \cdot F(X_U) < 0$ ¹. As $t \rightarrow \infty$, the trajectory dynamics collapse onto a measure-zero fractal attractor $I_U \subset X_U$

¹. The asymptotic statistical distribution on I_U is described by a Sinai-Ruelle-Bowen (SRB) measure μ_{SRB} , which is singular with respect to the Lebesgue measure on the embedding space X_U ¹.

Conversely, Bohmian mechanics and Madelung hydrodynamics are volume-preserving and conservative in probability space⁶. The Bohmian continuity equation

$\partial_t \rho_B + \nabla \cdot (\rho_B v_B) = 0$ strictly conserves the Born-rule probability measure $d\mu_B = \rho_B dx$ over time⁶.

Marginalization Mechanism

Reconciling a dissipative flow governed by an SRB measure with a conservative continuity equation requires decomposing the phase-space coordinates¹. Let the cosmological state space decompose into macro-spatial coordinates $x \in \mathbb{R}^3$ and unobservable internal cosmological sub-coordinates $y \in Y_U$:

$$X_U = \mathbb{R}^3 \times Y_U$$

While the full cosmological flow contracts phase-space volume in X_U along the internal Y_U dimensions (driving trajectories onto the fractal leaves of I_U), the projected marginal measure on configuration space $\mathbb{R}^3$:

$$\rho_B(x, t) = \int_{Y_U} d\mu_{\text{SRB}}(x, y, t)$$

remains non-singular and absolutely continuous with respect to the Lebesgue measure dx ¹. Because dissipation occurs orthogonal to spatial configuration space, the projected marginal velocity field $v_B(x, t)$ exhibits zero net probability sinkage in $\mathbb{R}^3$ ¹. Thus, the macroscopic conservation of probability is mathematically compatible with dissipative SRB contraction on I_U , provided the dissipation is confined to the internal p -adic dimensions¹.

Kinematic Emergence of the Quantum Potential Q

Information Geometry and Stochastic Origins

In Bohmian mechanics, particle acceleration is driven by both classical forces and the non-local quantum potential $Q(x, t)$ ⁷:

$$Q(x, t) = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} = -\frac{\hbar^2}{8m} \left(2 \frac{\nabla^2 \rho}{\rho} - \frac{|\nabla \rho|^2}{\rho^2} \right)$$

In information geometry, Q is the exact functional derivative of the spatial Fisher Information $I_F(\rho)$ ⁶.

$$I_F(\rho) = \int_{\mathbb{R}^3} \frac{|\nabla \rho(x)|^2}{\rho(x)} dx \implies \frac{\delta}{\delta \rho} \left(\frac{\hbar^2}{8m} I_F(\rho) \right) = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} = Q(x)$$

In Nelson's stochastic mechanics and Stochastic Quantum Hydrodynamics (SQHM), Q emerges kinematically from a diffusion process⁷. A particle subjected to forward and backward stochastic diffusions possesses a current velocity $v = \frac{\nabla S}{m}$ and an osmotic velocity $u =$ ⁸. The osmotic kinetic energy density of the ensemble matches the Fisher information⁶:

$$\frac{1}{2}m \int \rho |u|^2 dx = \frac{\hbar^2}{8m} \int \frac{|\nabla \rho|^2}{\rho} dx = \frac{\hbar^2}{8m} I_F(\rho)$$

Variation of this osmotic kinetic energy under mass-preserving flow generates Q ⁶.

Derivation from Microscopic Trajectory Fluctuations

In Invariant Set Theory, trajectory dynamics are non-stochastic and deterministic¹. However, the fractal p -adic geometry creates intrinsic spatial fluctuations between neighboring trajectory leaves on I_U ¹.

Let δx_p denote the spatial distance between two adjacent trajectory leaves on I_U at p -adic resolution p^{-N} ¹. The ensemble distribution of trajectory positions x across the fractal cross-section induces a spatial density profile $\rho(x)$ ¹. Because trajectories on I_U are bounded by the fractal invariant set's Cantor structure, local perturbations in density $\rho(x)$ alter the geometric spacing of neighboring trajectories according to $d_p(x, x + \delta x) \propto \rho(x)^{-1/d_F}$, where d_F is the Hausdorff dimension of I_U ¹.

Calculating the variance of spatial velocity fluctuations $\langle \delta v \otimes \delta v \rangle$ generated by these fractal trajectory bounds yields an effective osmotic drift velocity $u_{IST}(x) = \frac{\hbar}{2m} \nabla \ln \rho(x)$ ¹. Inserting this osmotic velocity into the kinetic momentum balance equation generates the osmotic stress tensor $\mathbb{P}_{ij}^{osm} = \rho u_i u_j$ ⁸. Taking its divergence yields:

$$\frac{1}{\rho} \nabla \cdot (\rho u_{\text{IST}} \otimes u_{\text{IST}}) + \text{higher-order spatial corrections} = \nabla Q$$

Thus, the quantum potential Q arises directly as the macroscopic pressure exerted by fractal trajectory dispersion on I_U when marginalized over configuration space $\mathbb{R}^3$.

Comparative Assessment Matrix

The theoretical status, mathematical mechanisms, and resolution pathways for the five adversarial claims regarding the coarse-graining of Invariant Set Theory into Bohmian hydrodynamics are structured in the following comparative analysis:

Adversarial Claim / Objection	Primary Mathematical Origin	Theoretical Finding / Status	Necessary Condition for Legitimate Bridge
Fractal Non-Rectifiability	Cheeger differentiability failure; total disconnectedness of $\mathbb{Z}_p$ [cite: 9, 12]	Verified Obstruction at Finite p : Weaver derivations vanish; $\text{Der}(I_U) = \quad$.	Require singular limit $p \rightarrow \infty$; differential calculus is entirely extrinsic ex post ¹ .
Kinetic Closure & Dispersion	Multistreaming, caustic formation, multi-valued phase spaces ¹⁸	Qualified Objection: Monostreaming fails ($\mathbb{P}_{ij} \neq 0$) due to trajectory crossing ¹⁸ .	Trajectory dispersion stress $\mathbb{P}_{ij}$ must match the quantum stress tensor $\mathbb{T}_{ii}^{\text{quant}}$ ⁸ .
Wallstrom Circulation	Multivalued phases; unquantized circulation on Ω_+ [cite: 6]	Resolved via Decomposition: Target smoothness fixes phase period $2\pi\hbar$.	IST rational phases $\mathbb{Q}_2(N)$ must satisfy global scalar descent to $\psi \in C^\infty(M, \mathbb{C})$ ⁴ .

SRB / Dissipation Contraction	Phase-space volume contraction $\nabla \cdot F < 0$ on attractors ¹	Resolved via Marginalization: Dissipation is confined to internal Y_U dimensions ¹ .	Projected spatial measure $\mu_{\text{SRB}} _{\mathbb{R}^3}$ must remain absolutely continuous ($\partial_t \rho + \nabla \cdot (\rho v) = 0$) ¹ .
Emergence of Quantum Potential Q	Non-local quantum kinetic energy generation ⁷	Verified Kinematic Origin: Q is the variational derivative of Fisher Information I_F ⁶ .	Microscopic fractal spatial fluctuations δx_p must generate osmotic drift $u = \dots$ ⁸ .

Synthesized Conclusions and Technical Assessment

Evaluating non-smooth analysis, kinetic transport theory, and quantum hydrodynamics leads to the conclusion that a direct, continuous spatial coarse-graining of Invariant Set Theory into the

Bohmian velocity field at finite p is mathematically impossible¹. Because I_U at finite p is homeomorphic to the p -adic integers $\mathbb{Z}_p$, it lacks non-trivial rectifiable curves¹.

Consequently, all Weaver metric derivations and continuous spatial vector fields on I_U vanish identically ($\text{Der}(I_U) \equiv \{0\}$), preventing the definition of intrinsic spatial velocity vectors on I_U prior to taking the continuum limit¹. Furthermore, multi-trajectory chaotic intersections on I_U generate non-zero velocity dispersion ($\mathbb{P}_{ij} \neq 0$), breaking classical monostreaming hydrodynamic closure¹⁸.

However, a mathematically legitimate bridge exists in the singular limit $p \rightarrow \infty$ combined with spatial coordinate marginalization, subject to three technical criteria¹:

- Singular Topological Convergence:** Macroscopic manifold structures and Bohmian vector fields must emerge as extrinsic constructions in the singular limit $p \rightarrow \infty$, where the p -adic metric d_p collapses and dyadic rational phases $\mathbb{Q}_2(N)$ become dense in $\mathbb{S}^1$.
- Topological Scalar Descent:** To evade Wallstrom's circulation objection, the

coarse-grained phase 1-form dS must satisfy the Reddiger-Poirier global scalar descent condition, guaranteeing that the hydrodynamic fields lift to a single-valued complex wavefunction $\psi \in C^\infty(M, \mathbb{C})$ with integer circulation $\oint v \cdot dx \in \frac{2\pi\hbar}{m} \mathbb{Z}$ ⁶.

3. **Stress Tensor Identification:** The kinetic velocity dispersion tensor $\mathbb{P}_{ij}$ generated by fractal trajectory fluctuations on I_U must re-emerge macroscopically as the quantum stress tensor $\mathbb{T}_{ij}^{\text{quant}} = -\frac{\hbar^2}{4m^2} \rho \partial_i \partial_j \ln \rho$, whose divergence produces the quantum potential $Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}$ via the variation of Fisher Information⁶.

When these three conditions are satisfied, the microscopic deterministic dynamics on Palmer's invariant set (I_U, T_t, μ_I) map onto macroscopic Bohmian hydrodynamics, establishing Bohmian mechanics as the effective continuum field theory of Invariant Set Theory¹.

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